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ODRL Policy Enforcement with ODRE

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About me

POLITÉCNICA

Research

Semantic Interoperability

Distributed Discovery

Policies for Data Usage

OPAL WORKSHOP

Standardization

ETSI

W3C

Alliance for IoT and Edge Computing Innovation

oneM2M

Web of Things (WoT) Discovery
W3C Recommendation 05 December 2023

ODRL Formal Semantics
Draft Community Group Report 13 April 2026

ODRE

WoTHive

EU Projects

DELTA

COGITO

STICS
Smart Transformation & Innovation Consortium Slovakia

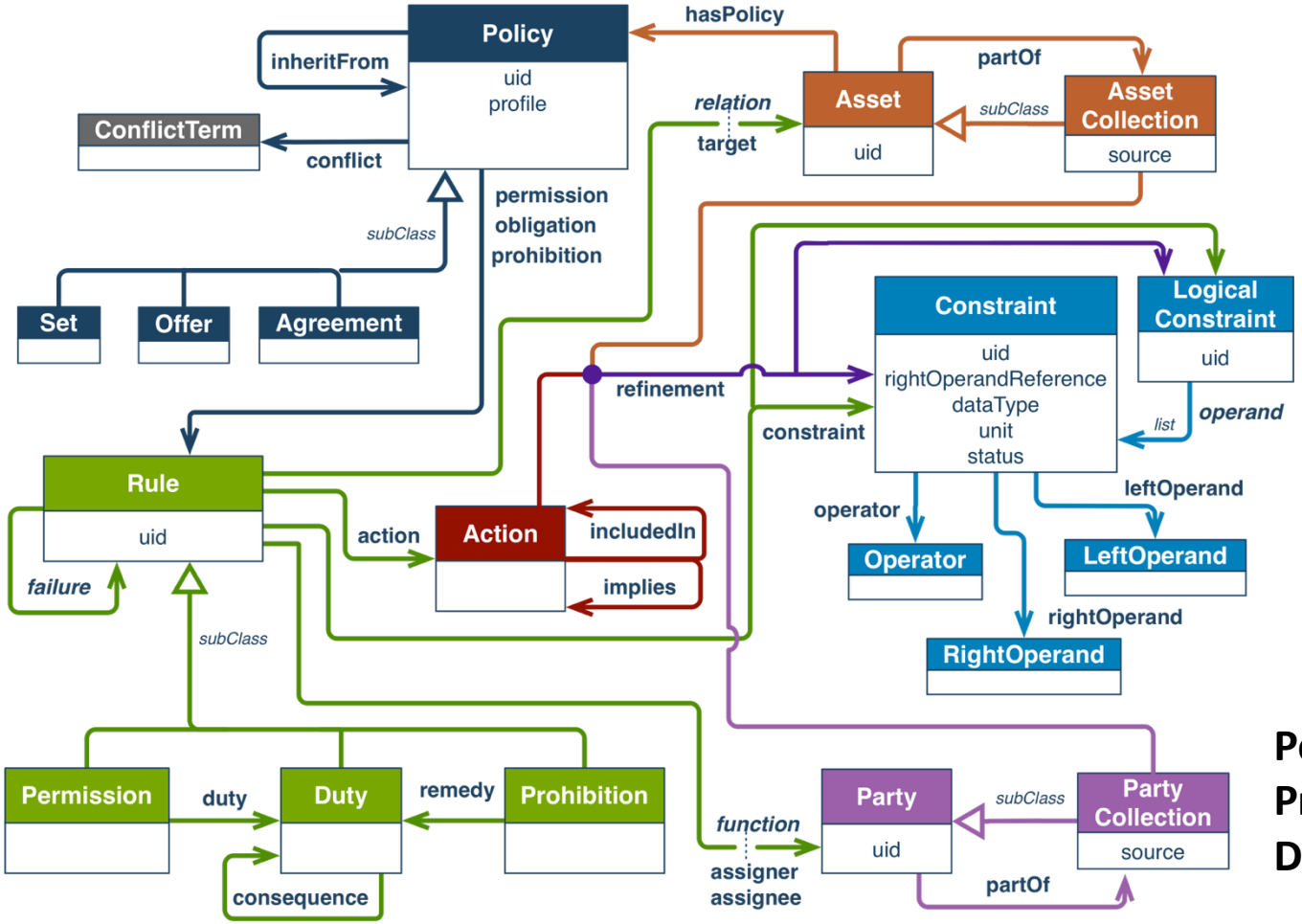
VICINITY 2020

BIMERR
RENOVATION 4.0

AURORAL

ODRL Information Model 2.2

W3C Recommendation 15 February 2018



Permission
Prohibition
Duty

ODRL Formal Semantics

Draft Community Group Report 13 April 2026

```
{
"@context": "http://www.w3.org/ns/odrl.jsonld",
"@type": "Set",
"uid": "http://example.com/policy/A1",
"permission": [{
   "@id": "http://example.com/rule/A1",
   "target": "http://example.com/document/1234",
   "assigner": "http://example.com/party/16",
   "action": "distribute",
   "constraint": [{
       "@id": "http://example.com/constraint/A1",
       "leftOperand": "dateTime",
       "operator": "lt",
       "rightOperand":  { "@value": "2018-01-01", "@type": "xsd:date" }
   }]
}]
}
```

ODRL Formal Semantics

Draft Community Group Report 13 April 2026

```
{
"@context": "http://www.w3.org/ns/odrl.jsonld",
"@type": "Set",
"uid": "http://example.com/policy/A1",
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   }]
}]
}
```

Resource protected

Policy owner

Action allowed

Conditions

ODRL Formal Semantics

Draft Community Group Report 13 April 2026

```
{
"@context": "http://www.w3.org/ns/odrl.jsonld",
"@type": "Set",
"uid": "http://example.com/policy/A1",
"permission": [{
```

Case	Policy	State of the World	Evaluation Request	Satisfaction of class, target, assignee	Satisfaction state of `constraint:A1`	Activation state of the `permission`	Control state of the `permission`	Evaluation Report
A1-1	policy:A1	empty	party:1 distributes document:1234 on 2017-12-19T15:00:00	satisfied	satisfied	active	permit the action	report
A1-2	policy:A1	empty	party:1 distributes document:1234 on 2019-12-19T15:00:00	satisfied	not-satisfied	inactive	deny the action	report

```
    }]
}]
}
```

https://github.com/orgs/ODRE-Framework

- Native ODRL support
- ODRL extensions
- Code-base functionalities (interopolation)

ODRE in a nutshell: policy template and variables

1. ODRL TEMPLATE (JSON-LD)
policy-template.jsonld

```
{
  "@context": "http://www.w3.org/ns/odrl.jsonld",
  "@type": "Offer",
  "uid": "http://example.com/policy:6163",
  "profile": "http://example.com/odrl:profile:10",
  "permission": [{
    "target": "http://example.com/document:1234",
    "assigner": "http://example.com/org:616",
    "action": "distribute",
    "constraint": [{
      "leftOperand": "dateTime",
      "operator": "lt",
      "rightOperand": { "@value": "2025-01-01",
                        "@type": "xsd:date" }
    },{
      "leftOperand": { "@value": "{{retrieve_token()}}",
                       "@type": "xsd:string" },
      "operator": "eq",
      "rightOperand": { "@value": "{{token}}",
                        "@type": "xsd:string" }
    }]
  }]
}
```

This is the ODRL template written in JSON-LD.
It contains placeholders (e.g., {{retrieve_token()}}, {{token}}) that will be replaced with values from the evaluation payload.

2. EVALUATION PAYLOAD (PYTHON)
evaluation_payload.py

```python
# Evaluation payload as Python variables
 evaluation_payload = {
     "token": token,
     "retrieve_token()": retrieve_token
}

# Data used for instantiating the template
token = "abc123xyz"

def retrieve_token() -> str:
    """
    Returns a token string.
    """
    ...  # Implementation not shown
```

ODRE INTERPOLATOR

The ODRE Interpolator replaces placeholders in the template with values from the evaluation payload and produces the final ODRL policy.

3. ODRL POLICY (JSON-LD)
odrl-policy.html

```
{
  "@context": "http://www.w3.org/ns/odrl.jsonld",
  "@type": "Offer",
  "uid": "http://example.com/policy:6163",
  "profile": "http://example.com/odrl:profile:10",
  "permission": [{
    "target": "http://example.com/document:1234",
    "assigner": "http://example.com/org:616",
    "action": "distribute",
    "constraint": [{
      "leftOperand": "dateTime",
      "operator": "lt",
      "rightOperand": { "@value": "2025-01-01",
                        "@type": "xsd:date" }
    },{
      "leftOperand": { "@value": "abc123xyz",
                       "@type": "xsd:string" },
      "operator": "eq",
      "rightOperand": { "@value": "abc123xyz",
                        "@type": "xsd:string" }
    }]
  }]
}
```

Result: a valid ODRL policy in JSON-LD (HTML serialization), with all placeholders replaced by the evaluation payload data.

ODRL Template (JSON-LD) — Structure with placeholders
+
Evaluation Payload (Python) — Values to instantiate the template
→
ODRL Policy (HTML) — Instantiated policy document

ODRE in a nutshell: interpolation

```
policy = """
{
    "@context": "http://www.w3.org/ns/odrl.jsonld",
    "@type": "Offer",
    "uid": "http://example.com/policy:6163",
    "profile": "http://example.com/odrl:profile:10",
    "permission": [{
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       "assigner": "http://example.com/org:616",
       "action": "distribute",
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           "operator": "lt",
           "rightOperand":  { "@value": "2025-01-01", "@type": "xsd:date" }
       },{
           "leftOperand": { "@value": "{{retrieve_token()}}" , "@type": "xsd:st
           "operator": "eq",
           "rightOperand":  { "@value": "{{token}}", "@type": "xsd:string" }
       }]
   }]
}
"""
```

ODRE in a nutshell: interpolation

```
policy = """
```

```python
def get_token():
    return "AAA"

usage_decision = ODRE().set_debug(True).enforce(policy, interpolations={'token' : 'AAA',
'retrieve_token' : get_token})
print(usage_decision)
```

```
        "constraint": [{
            "leftOperand": "dateTime",
            "operator": "lt",
            "rightOperand":  { "@value": "2025-01-01", "@type": "xsd:date" }
        },{
            "leftOperand": { "@value": "{{retrieve_token()}}" , "@type": "xsd:str
            "operator": "eq",
            "rightOperand":  { "@value": "{{token}}", "@type": "xsd:string" }
        }]
    }]
}
"""
```

ODRE in a nutshell: interpolation

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policy = """
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ODRE in a nutshell: interpolation

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        "constraint": [{
            "leftOperand": "dateTime",
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            "rightOperand":  { "@value": "2025-01-01", "@type": "xsd:date" }
        },{
            "leftOperand": { "@value": "{{retrieve_token()}}" , "@type": "xsd:st
            "operator": "eq",
            "rightOperand":  { "@value": "{{token}}", "@type": "xsd:string" }
        }]
    }]
}
"""
```

ODRE in a nutshell: policy compilation

1. INPUT: ODRL POLICY (JSON-LD)
odrl-policy.jsonld

```json
{
 "@context": "http://www.w3.org/ns/odrl.jsonld",
 "@type": "Offer",
 "uid": "http://example.com/policy:6163",
 "profile": "http://example.com/odrl:profile:10",
 "permission": [
   {
    "target": "http://example.com/document:1234",
    "assigner": "http://example.com/org:616",
    "action": "distribute",
    "constraint": [
      {
         "leftOperand": "dateTime",
         "operator": "lt",
         "rightOperand": {
             "@value": "2018-01-01",
             "@type": "xsd:date"
         }
      }
    ]
 ]
}
```

This ODRL policy grants permission to distribute the target resource only if the current dateTime is before 2018-01-01.

ODRE COMPILER

The ODRE Compiler analyzes the ODRL policy, extracts the constraints, and generates a Python script that evaluates them and returns True or False.

2. OUTPUT: PYTHON CONSTRAINT SCRIPT
policy_check.py

```python
from datetime import datetime

def odre_lt(left, right):
    """Return True if left < right, else False."""
    return left < right

def odre_datetime():
    """Return the current date and time."""
    return datetime.now().isoformat(timespec="seconds")

def evaluate():
    """Evaluate the policy constraint:
       current dateTime < 2018-01-01T00:40:30."""
    return odre_lt(odre_datetime(), "2018-01-01T00:40:30")

if __name__ == "__main__":
    print(evaluate()) # Prints True or False
```

The generated Python script evaluates the constraint from the policy and returns True if the current dateTime is before 2018-01-01T00:40:30, otherwise False.

ODRL Policy (JSON-LD)
Input policy with constraints

→

ODRE Compiler
Extracts constraints and generates Python script

→

Python Script
Evaluates constraints and returns True or False

Conclusions

- Policy templates:
  - Are not always correct RDF
  - Allow to “generate” on the fly a large number of policies (on demand or beforehand)
  - May contain complex logic
- Policy compilation:
  - Requires bounding ontological terms to code (e.g., Python or Java)

https://colab.research.google.com/drive/10XHbTmSKtOJlYmg3f94vZDyIR8Txvvi5?usp=sharing

Thank you!